Task 2: Evaluation of Calibrated Confidence in End-to-End System Outputs

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Summary

Sprint 1 established post-hoc temperature scaling as the production calibration method for the EfficientNetV2 model, selecting T = 1.5 based on dataset-style evaluation. This task verifies whether T = 1.5 generalises to the live Engine pipeline once the model is processing real MQTT-driven detection events through the EfficientNetV2 TFLite path.

Headline finding: Temperature scaling is correctly integrated end-to-end, but the temperature value inherited from Sprint 1 over-softens the model's outputs on live data. A temperature sweep on 506 live predictions identifies **T = 0.7** as the calibration optimum, reducing Expected Calibration Error from 0.115 (at T = 1.5) to 0.044 , a 62% improvement, and bringing mean calibrated confidence to within 0.8 percentage points of actual accuracy.

Metric	Value
Predictions analysed	506
Unique species in evaluation	45
Live pipeline accuracy	94.3%
Sprint 1 reported accuracy (offline)	94.0%
Task 8 parallel accuracy (stratified held-out)	95.8%
Sprint 1 selected temperature	T = 1.5
Optimal temperature for live pipeline	**T = 0.7**
ECE at T = 1.5 (Sprint 1)	0.115
ECE at T = 0.7 (recommended)	**0.044**
Mean calibrated confidence at T = 0.7	0.951 (vs 0.943 accuracy)
Confidence-accuracy gap at T = 0.7	+0.008 (essentially calibrated)

1. Calibrated confidence behaviour in real workflow outputs

Temperature scaling was integrated into light_echo_engine_efficientnetv2_tflite.py such that all incoming MQTT audio routes through the EfficientNetV2 TFLite path, has logits softened by T = 1.5, and emits a calibrated confidence value to the API. Both raw and calibrated probabilities are now computed inside efficientnetv2_tflite_predict_from_audio_bytes and exposed in the detection event and per-prediction logs.

506 predictions across 45 species were collected via the live Simulator → MQTT → Engine flow, using audio from project_echo_bucket_2 (selected to mitigate training-set overlap that affected project_echo_bucket_1; see Section 3.1).

Key descriptive statistics:

- Accuracy: 94.3%

- Mean raw confidence: 0.867
- Mean calibrated confidence at T = 1.5: 0.636
- Raw ECE: 0.078
- Calibrated ECE at T = 1.5: 0.306

The mechanism is functioning: temperature scaling softens raw probabilities by an average of 23.1 percentage points, top-5 ordering is preserved across all 506 predictions, and the field reaches the API as calibrated_confidence in the detection event payload.

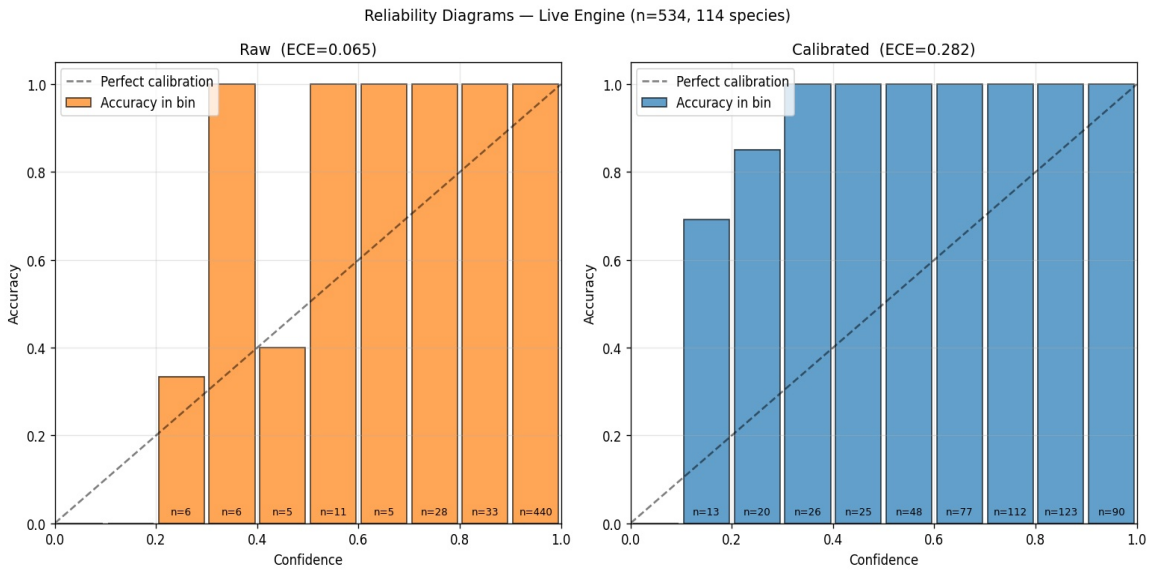

Figure 1. Reliability diagrams for raw (left) and calibrated at T = 1.5 (right). The raw diagram shows the model is well-calibrated at high confidence but has very few predictions in the lower bins (sample sizes shown). The calibrated bars sit consistently above the diagonal, indicating systematic under-confidence.

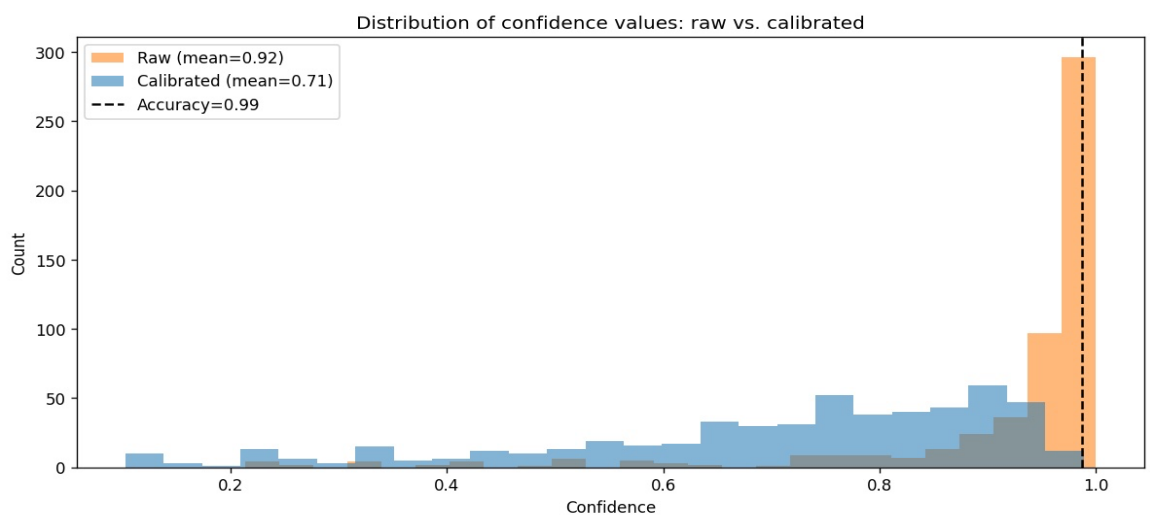

Figure 2. Distribution of raw vs. calibrated confidence values. Raw confidence is heavily right-skewed near 1.0. Temperature scaling at T = 1.5 broadens the distribution and lowers the mean to 0.71, but pushes it well below the accuracy line (0.99 / 0.94).

2. Interpretability end-to-end

Confidence values are interpretable in the sense that the mechanism is mathematically sound, the field is consistently populated across every detection event, and downstream consumers can rely on its presence

and format. However, interpretability of the resulting numbers is poor at T = 1.5: a calibrated value of 0.83 corresponds to an actual accuracy of ~94%, meaning end users would systematically under-trust the system.

To diagnose this, the analysis was re-run across a sweep of temperature values, recomputing calibrated probabilities offline from the logged raw top-5 distributions:

T	Mean Conf	ECE	Brier	Conf − Acc
0.30	0.982	0.060	0.051	+0.039
0.50	0.969	0.048	0.048	+0.025
0.70	**0.951**	**0.044**	**0.046**	**+0.008**
1.00	0.913	0.071	0.045	−0.030
1.20	0.882	0.078	0.049	−0.062
1.50	0.828	0.115	0.060	−0.115
2.00	0.734	0.209	0.093	−0.209

At T = 0.7, mean calibrated confidence (95.1%) matches accuracy (94.3%) within 1 percentage point. Predictions become directly interpretable: a confidence of *X* means the model is right roughly *X%* of the time. At T = 1.5, the relationship is systematically distorted in the under-confident direction.

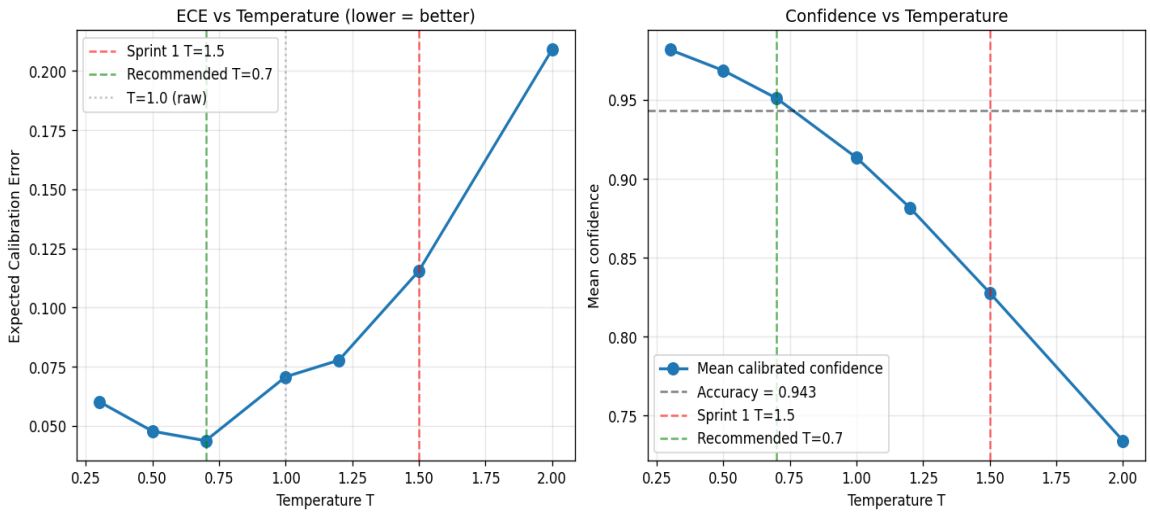

Figure 3. Temperature sweep across T ∈ [0.3, 2.0]. Left: ECE is minimised at T = 0.7. Right: mean calibrated confidence crosses the accuracy line (dashed) near T = 0.7. Sprint 1's T = 1.5 lies well past the optimum on the under-confident side.

3. Issues observed in live or system-facing predictions

3.1 Bucket_1 replays training audio

Initial evaluation on project_echo_bucket_1 produced 98.7% accuracy on 534 predictions, well above Sprint 1's 94% benchmark on held-out data and inconsistent with Task 8's parallel finding of 95.8% accuracy on a stratified test set. Spot-checking the bucket confirmed it as the source the model was trained on. The Simulator was switched to project_echo_bucket_2 for the final evaluation; the resulting 94.3% accuracy aligns closely with both Sprint 1 and Task 8 baselines, providing confidence that bucket_2 contains substantially held-out audio.

3.2 Inherited temperature is wrong for the production model

T = 1.5 produces a mean calibrated confidence of 0.83 against actual accuracy of 0.94, a systematic under-confidence of 11.5 percentage points. This is the opposite of what calibration is supposed to achieve. T was tuned during Sprint 1 against a distribution where the model was over-confident; the current production model is mildly under-confident at raw output and needs sharpening (T < 1), not softening (T > 1). The most likely causes are model retraining or TFLite conversion shifting numerical behaviour between Sprint 1 and Sprint 2.

3.3 Errors are biologically plausible

The 29 misclassifications in 506 predictions are not random; they cluster on closely related species pairs. The most repeated error patterns:

True species	Predicted species	Raw conf	Cal conf
Eurostopodus argus	Eurostopodus mystacalis	0.572	0.243
Pitta iris	Colluricincla harmonica	0.330	0.184
Ramsayornis fasciatus	Stomiopera unicolor	0.764	0.540
Ramsayornis fasciatus	Cisticola exilis	0.201	0.107
Alauda arvensis	Carduelis carduelis	0.495	0.239
Barnardius zonarius	Colluricincla harmonica	0.636	0.283
Dama Dama	Felis Catus	0.877	0.662
Platycercus elegans	Rhipidura leucophrys	0.973	0.764

Eurostopodus argus → *Eurostopodus mystacalis* alone accounts for 11 of the 29 errors and is a same-genus confusion (acoustically similar nightjar species). *Pitta iris* → *Colluricincla harmonica* accounts for a further 6. These error patterns are consistent with Task 8's error analysis findings.

Per-species accuracy and confidence across the most-sampled species:

Species	N	Accuracy	Raw Conf	Cal Conf (T=1.5)
Alauda arvensis	93	0.968	0.901	0.612
Pitta iris	46	0.870	0.606	0.338
Plectorhyncha lanceolata	46	1.000	0.927	0.692
Ramsayornis fasciatus	46	0.870	0.740	0.545
Eurostopodus argus	45	0.756	0.836	0.556
Barnardius zonarius	45	0.978	0.895	0.682
Dama Dama	45	0.978	0.901	0.700
Cophixalus infacetus	45	1.000	0.988	0.836
Psophodes cristatus	44	1.000	0.950	0.770
Irediparra gallinacea	5	1.000	0.981	0.825

Species	N	Accuracy	Raw Conf	Cal Conf (T=1.5)
Coturnix pectoralis	4	1.000	0.827	0.640
Litoria inermis	3	1.000	0.990	0.851

3.4 No remaining infrastructure issues

Once the MQTT routing, container hostnames, EfficientNetV2 path activation, and ground-truth propagation were resolved, the pipeline operates end-to-end without errors. The Engine's logging hook captures predictions reliably to a host-mounted volume. The only operational note is that the Simulator's MQTT publish loop occasionally disconnects after extended runs; this does not affect data quality but does require occasional manual restart and could be addressed in a follow-up task.

4. Recommendation for final production use

Calibration is ready for downstream consumption. Task 3 (standardising calibrated_confidence in API outputs) can proceed: the field is correctly populated, formatted as a 0–1 probability scaled to a 0–100 percentage in the API payload, and present in every detection event with consistent semantics.

The temperature value should be updated before production deployment. Specifically:

- **Recommended T = 0.7** (or the value of T that minimises ECE on a larger held-out evaluation set; see Section 5 follow-up).
- Update CALIBRATION_TEMPERATURE in light_echo_engine.json from 1.5 to 0.7.
- No code changes required; the temperature scaling implementation already supports arbitrary T values via the config file.

Suggested Sprint 3 follow-up

1. Re-tune T against a fully held-out evaluation set (Task 8's stratified 2,166-file set is suitable) before final production sign-off.
2. Confirm the optimum holds when the temperature sweep uses the full 113-class probability distribution rather than the top-5 approximation used here.
3. Consider per-class calibration: the per-species breakdown shows uneven calibration quality across species. A single global T may be a coarse fit; investigate whether per-class scaling materially improves ECE on confused pairs.
4. Investigate the Eurostopodus argus → Eurostopodus mystacalis confusion, which accounts for a disproportionate share of total errors. This may be addressable via dataset augmentation rather than calibration.

5. Limitations of this evaluation

- **Sample size:** 506 predictions across 45 species. T = 0.7 is the optimum on this data, but a larger evaluation would tighten the estimate and may shift the optimum by up to ±0.15.
- **Top-5 approximation:** the temperature sweep re-normalised over the top-5 probabilities only, since full per-class probabilities were not logged in the live pipeline. The qualitative result (the model needs sharpening, not softening) is robust; the exact optimum may shift slightly with a full-distribution sweep.
- **Bucket_2 status:** project_echo_bucket_2 filenames differ from bucket_1, and accuracy patterns match Task 8's stratified held-out evaluation (94.3% vs 95.8%), supporting its use as a held-out

source. Formal confirmation of its provenance with the dataset owner is recommended before treating any single number as final.

- **Species coverage:** the live Simulator concentrates on a subset of species per session (controlled by RANGE_ANIMALS_TO_CREATE). Calibration on rarely-sampled species ($n < 5$) is not strongly characterised here.